Walking Speed Classification for ECE 591 Final Project

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***Abstract*—This paper presents a novel approach to classifying walking speed based on foot positions, employing a 2D Convolutional Neural Network (CNN) to process data from 24 position markers on both feet. Gait speed is an evaluative statistic in a variety of fields, ranging from athletics to physical therapy. The model is trained and evaluated using a dataset from Kaggle, which contains data from 21 subjects walking at various speeds. This paper examines the limitations of built-in classification functions, and discusses the comparative benefits of the custom model that includes a CNN architecture with features such as categorical crossentropy loss, Adam optimizer, and 120 epochs. The developed model achieves 94% accuracy for training and a 95% accuracy for validation, values which are significantly higher than those found in the built-in functions. While these accuracy values are good, we also propose additional steps that could be used to improve upon our model in future iterations.**

I. Introduction

Gait speed is an important physical characteristic in a variety of fields. In athletics, it can be used to facilitate training regimens and evaluate or rank individual performance. In physical therapy for older adults, it can be an indicator of a patient's level of mobility. Especially in older adults, walking speed often correlates with the individual's overall health. Increasing walking speed is often a goal in these types of scenarios, but before trying to alter the walking speed on an individual, there must be a robust method of classifying different walking speeds.

In this paper, the positions of the foot are used to classify walking speed into 12 different categories. Foot position is determined by examining 4 data markers on each foot. Each sample is then fed into a 2D Convolutional Neural Network (CNN) to extract features pertaining to the relationship of each marker with another, and classify that sample into one of the twelve speed classifications.

II. Related Work

Hoangm, et all implemented a combination of 2D and 1D CNN as a means to detect Parkinson's Disease in subjects [1]. This study used Vertical Ground Reaction Force (VGRF) data from 93 Parkinson's patients and 73 healthy control subjects. Each subject walked for 2 minutes at a self-selected pace while wearing force sensitive insoles in their shoes. Each insole took VGRF measurements at 8 locations at a sampling rate of 100Hz. This data was preprocessed into 16x240 images, each representing 2.4 seconds of walking data. These images were then passed through the model shown in Figure 1.

The authors of this paper chose to combine 2D and 1D convolution because, according to them, each one has its own specific merits. 2D Convolution is better at extracting the way in which forces at different points relate to each other at points in time. However, since each image represents 2.4 seconds of data, not just a single point in time, 1D Convolution is used as well. 1D Convolution is better at extracting features from time series data. The combination of the 2 types of convolution produced an accuracy 88.7%.

Layer (type)	Output Shape	Param #
conv2d_6 (Conv2D)	(None, 16, 240, 24)	1200
activation_9 (Activation)	(None, 16, 240, 24)	0
max_pooling2d_6 (MaxPooling2	(None, 8, 120, 24)	0
dropout_6 (Dropout)	(None, 8, 120, 24)	0
reshape_6 (Reshape)	(None, 960, 24)	0
conv1d_2 (Conv1D)	(None, 957, 24)	2328
flatten_2 (Flatten)	(None, 22968)	0
dense_4 (Dense)	(None, 2)	45938
activation_10 (Activation)	(None, 2)	0

Total params: 49,466
Trainable params: 49,466
Non-trainable params: 0

Fig. 1. Combined 2D and 1D CNN Model Summary

Shrestha, et all used a 2D CNN to classify walking speeds from vertical and horizontal components of accelerometer and gyroscope data from a smartphone [2]. 10 subjects were asked to walk on a treadmill at 5 different speeds ranging from 1mph to 3mph, with increments of 0.5mph. Subjects walked at each speed for 5 minutes while holding the smartphone in their hand, and both the acceleration and gyroscope data was sampled at a rate of 100Hz. The data was then preprocessed into 45x4 image files, with each row of the image corresponding to data from one of the four datasets previously mentioned. Four 2D Convolution steps were used with 16, 32, 48, and 64 filters respectively to classify the walking speed into 5 categories. Figure 2 provides a summary of the model steps.

Su et al. [3] employed Deep Convolutional Neural Networks (DCNN) for segmenting gait cycle recognition into five phases of classification across various walking speeds. The model utilized Inertial Measurement Unit (IMU) data, converting it into a 7x9x1 image size as input, and applied a normalization method using standard deviation. The training-test split was set at 70% for training and 30% for testing or 11 subjects for training and 1 subject for testing. The training involved 100

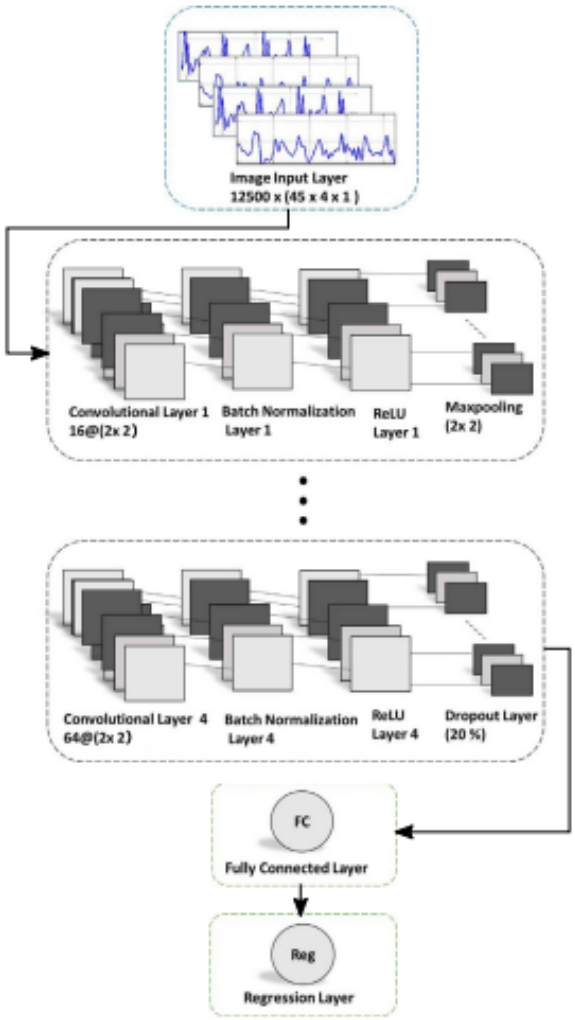


Fig. 2. Flowchart of 2D Convolutional Steps for Smartphone Data Model

epochs with early stopping, and the evaluation metric used was accuracy. Ultimately, their model achieved an accuracy of 97% during an offline evaluation of gait phase recognition, with the highest accuracy observed in the swing phase and the lowest in terminal stance. They transformed the input data from 1D to 2D, treating it as image data, a technique that will also be employed in our model.

Moura et al. [4] employed a single foot-mounted Inertial Measurement Unit (IMU). They utilized a shallow Convolutional Neural Network-Long Short-Term Memory (CNN-LSTM) architecture to classify activities on unstructured ground, including stair and level ground, with healthy subjects performing activities such as ascending and descending stairs. The combination of LSTM allows capturing past features, while CNN is adept at extracting relevant features. The inputs to the model consisted of accelerations and angular velocity. The implementation was carried out using Keras, with the RMSprop optimizer and cross-entropy loss function. The training process involved a batch size of 32 samples and was conducted over 50 epochs. Notably, training was performed on each subject, and validation was conducted with the same subject, contributing to a comprehensive and subject-specific evaluation. Prasanna et al. [5] used the KNN ML classifier to recognize irregularities in terrain for uneven types between tiles, grass, and stone. They utilized data from foot pressure sensors on four able-bodied subjects, with 16 optoelectronics embedded in the insole. The accuracy of their model in recognizing the type of uneven terrain on which subjects were walking was 87.5%, and they used confusion matrices to compare the final results for each class and analyze the differences and challenges associated with each.

III. Dataset and Features

The dataset used to train this model came from Kaggle [6]. It included 21 subjects, 10 males and 11 females. Each subject walked in a laboratory setting at 12 different speeds ranging from 0.6 m/s to 1.7 m/s, increasing at intervals of 0.1 m/s. Subjects walked at each speed for 1 minute. Figure 3 shows the setup of the lab that was used for these trials.

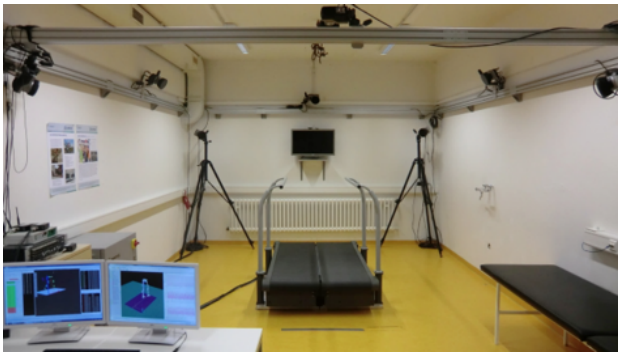


Fig. 3. Lab Setup for Data Collection [6]

2 types of data were collected during these trials: Ground Reaction Force (GRF) and Marker Positions. The three-dimensional GRF was recorded for each foot at a sampling rate of 1000Hz, and the three-dimensional position data was recorded at a sampling rate of 200Hz. 4 markers were placed on each foot, as shown in Figure 4, for a total of 24 marker data points per recorded sample.



Fig. 4. Marker Positions on Each Foot [6]

IV. Methods

After preparing the data, the classifier functions were employed to evaluate the results and performance. The *sklearn*

library was utilized for this purpose, incorporating the Gaussian Naive Bayes, Decision Tree, Modified SVM, and K-Neighbour algorithms. In the case of a small dataset, the classifiers exhibited high accuracy; however, when a larger dataset was employed, their individual accuracy proved to be unsatisfactory, as outlined in Table I. These findings indicate the necessity of developing a more sophisticated model to achieve an accuracy surpassing 73%.

TABLE I
UNACCEPTED ACCURACY OF CLASSIFIERS

Guassian NB (Naive Bayes)	37%
Decision Tree Classifier	38%
Modified SVM	56%
K-Neighbor	73%

One-hot encoding is a method used to represent categorical data as binary vectors. Each category is assigned a unique binary code, where only one bit is set to 1 in the corresponding column. This binary matrix format is commonly employed in machine learning, especially in deep learning models, to handle categorical labels[7]. In this project, one-hot encoding is applied to transform the output data into binary values due to the presence of 12 labels in the output.

Concerning the input data, the CNN model can adopt various approaches. The initial method involves inputting the data as 1D and subsequently flattening it for the model. Alternatively, another technique is employed to convert the input data into a 2D format, aligning with the approach commonly used for images in CNN. Consequently, the original 24 variables for each row are transformed into a 4 by 6 variable grid in 2D, and the corresponding techniques for 2D CNN are employed during training. This model was chosen over the 1D CNN because the input data was from individual points in time, not over the course of a time series.

The CNN model is configured with the following features: *Categorical_Crossentropy* as the loss function, *accuracy* as the evaluation metric, *Adam optimizer* for adaptive learning rates, starting at 0.001 with a minimum of 1e-6. A batch size of 64 enhances computational efficiency, and training spans 120 epochs, ensuring comprehensive learning from the dataset. These features collectively contribute to the model's effectiveness in multiclass classification. The details are also described in Table II.

The rationale for choosing *Categorical_Crossentropy*, as per the Keras library description, lies in its suitability for scenarios with two or more label classes, assuming the labels are provided in a one-hot representation. Alternatively, if integer-based labels are preferred, one may consider using the *SparseCategoricalCrossentropy* loss [8].

With all the above-mentioned techniques for preparing both the input and output data, a CNN model implemented consisting of a convolutional layer with 32 filters, followed by max-pooling. The data is then flattened and processed through a dense layer with 128 neurons using ReLU activation. A dropout layer helps prevent overfitting, and the final dense layer with softmax activation is designed for multi-class classification with 12 classes. The model details, including the number of parameters, are described in Table II.

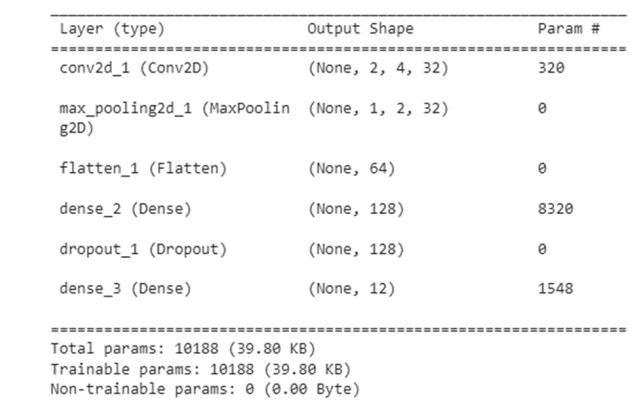


Fig. 5. Proposed Model Summary

TABLE II
MODEL FEATURES

Loss function type	Categorical_crossentropy
Metrics	accuracy
Optimizer	Adam
Learning Rate	0.001, adaptive with min 1e-6
Batch size	64
Epochs	12

V. RESULTS AND DISCUSSION

The entire process of training and compiling the system took approximately 35 minutes. In Figure 6, the results of the last 5 epochs are depicted, with the final accuracy for training and validation reaching 94% and 95%, respectively.

Additionally, in Figure 7, the loss values for both training and validation are displayed, demonstrating acceptable convergence for the loss function in both cases. Notably, there are two noticeable jumps at epochs 82 and 113. It is suggested that with more than 120 epochs, additional jumps might occur, albeit with smaller differences. However, it is crucial to consider the trade-off between accuracy and the number of epochs. Choosing a larger epoch value increases computation costs, and beyond a certain point, the accuracy improvement might not justify the extended computational time. To illustrate, with 120 epochs, a 95% accuracy is achieved, and potentially, with 400 epochs, accuracy may increase to 96%. However, the additional 6 hours of computation time make this increase arguably not worthwhile.

On a different note, Figure 8 illustrates the accuracy results, showcasing a commendable coverage of up to 95% for both training and validation. Nevertheless, a notable concern arises as the validation accuracy surpasses the training accuracy, potentially indicating overfitting or insufficient data variation. Given that the data in the paper solely comprises foot segment positions, incorporating additional variables such as force might mitigate this overfitting issue. This assumption stems from the observation that despite the use of dropout and augmentation in the project, overfitting was not effectively prevented.

Finally, Fig 9 visualizes the changes in the learning rate. It is evident that adapting the learning rate to the computational complexity by choosing lower values yields a nearly constant

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4725/4725 [==============================] - 17s 4ms/step - loss: 0.1235 - accuracy: 0.9493 - val_loss: 0.1022 - val_accuracy: 0.9581 - lr: 5.0000e-04
Epoch 116/120
4725/4725 [==============================] - 19s 4ms/step - loss: 0.1137 - accuracy: 0.9539 - val_loss: 0.0987 - val_accuracy: 0.9592 - lr: 2.5000e-04
Epoch 117/120
4725/4725 [==============================] - 17s 4ms/step - loss: 0.1139 - accuracy: 0.9538 - val_loss: 0.0985 - val_accuracy: 0.9583 - lr: 2.5000e-04
Epoch 118/120
4725/4725 [==============================] - 18s 4ms/step - loss: 0.1135 - accuracy: 0.9541 - val_loss: 0.0956 - val_accuracy: 0.9599 - lr: 2.5000e-04
Epoch 119/120
4725/4725 [==============================] - 17s 4ms/step - loss: 0.1133 - accuracy: 0.9545 - val_loss: 0.1035 - val_accuracy: 0.9565 - lr: 2.5000e-04
Epoch 120/120
4725/4725 [==============================] - 17s 4ms/step - loss: 0.1127 - accuracy: 0.9544 - val_loss: 0.0993 - val_accuracy: 0.9573 - lr: 2.5000e-04

Duration: 0:35:22.946152
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Fig. 6. Results of final 5 epochs

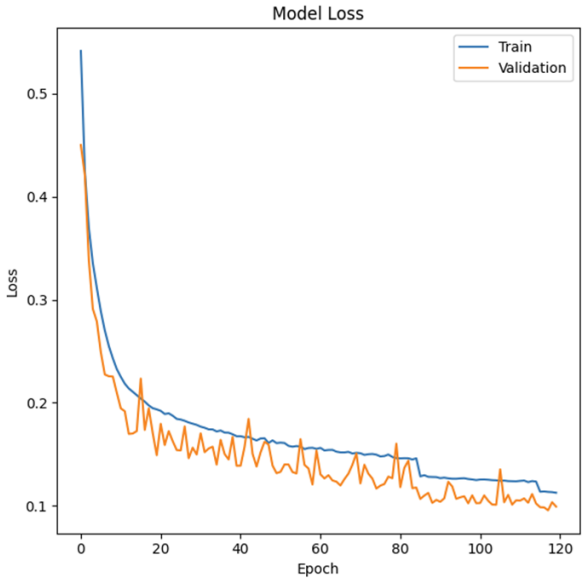


Fig. 7. Model loss

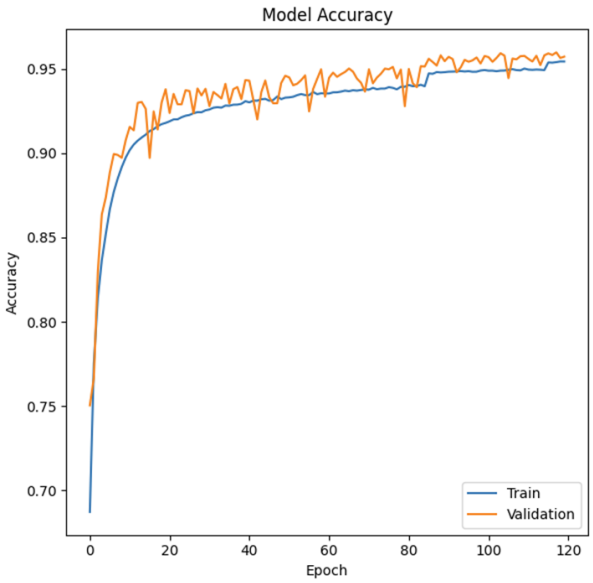


Fig. 8. Model accuracy

coverage in both loss and accuracy for both training and validation data.

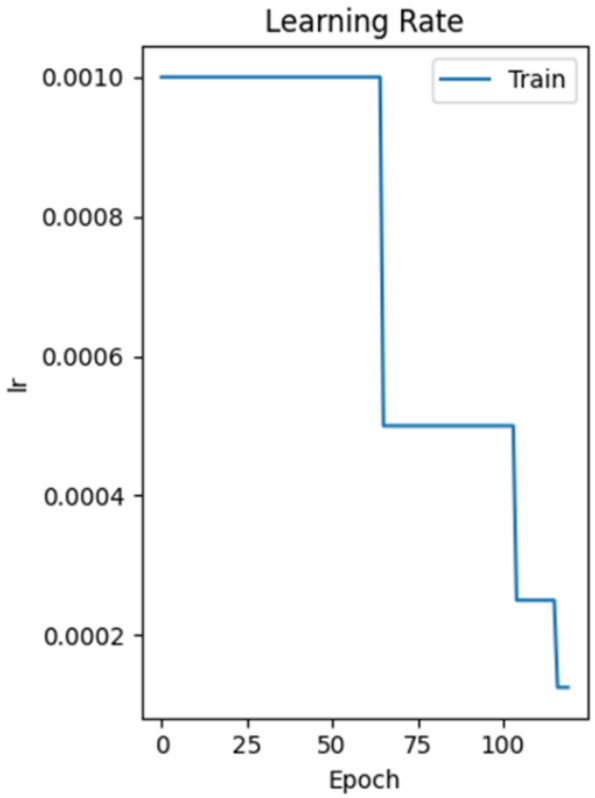


Fig. 9. Learning rate change

VI. Conclusion and Future Work

In this study, a Convolutional Neural Network (CNN) model is employed to recognize the walking speed of various subjects. The classification task involved 12 distinct walking speeds, with the model tasked to predict them based on the three components of the positional coordination of segments from each foot during walking. Remarkably, our model achieved an accuracy of 95%, demonstrating its efficacy in accurately detecting walking speeds. Additionally, the model exhibited acceptable coverage in calculating the loss function, indicating its ability to generalize and predict the behavior of provided data, both in the training and validation phases.

A pivotal technique employed in this study was the transformation of data dimensions from 1D to 2D, treating them

as image data. This innovative approach yielded noteworthy results, elevating the accuracy from 73% for the 1D version to 95% for the 2D version. The utilization of this technique not only showcased its effectiveness but also highlighted the potential of leveraging unconventional data representations to enhance model performance.

Our findings underscore the robustness of the CNN model in accurately predicting walking speeds, and the strategic use of dimensionality transformation emerges as a valuable technique for improving the model's overall performance. These results contribute to the evolving landscape of deep learning applications in gait analysis and hold promise for future advancements in understanding and predicting human movement patterns.

In the future, there are several modifications that could be made to the model to increase its classification accuracy. The accuracy of the current model is limited to data from four subjects. One cause of this could be the lack of significant difference between the data from the 12 different classes, as the difference in speed between consecutive classes is only 0.1m/s. To achieve higher accuracy over a larger set of data, it could be beneficial to categorize speed ranges in the form of "slow," "medium," and "fast" instead of distinct speeds so that there are fewer classes to categorize. This would still allow the model to make meaningful categorizations without the need to indicate an exact speed.

Additionally, data could be examined over time intervals instead of at finite points in time. Many of the models discussed in the "Related Work" section achieve higher accuracy by implementing this type of analysis. In fact, we spent significant time ourselves trying to develop a model that could examine data over a specific time range (for example, 1 second) to perform speed classification, but we were not able to produce a model that achieved an adequate level of accuracy before the submission deadline. As such, this could be a worthwhile model to revisit in the future.

Altering the data type could also provide benefits to future work. The dataset we used contained both position and force data, but we chose to only include position data in our model. These data were collected in a lab with specialized equipment, but in the real world, collecting data from position markers can be significantly challenging. Ground Reaction Force data, however, can be relatively simple to collect, with a force sensitive insole being a common tool used to collect this data. Categorizing the walking speed using this data could allow our model to be implemented in a wider range of areas, and would allow for speed classification to be conducted for longer periods of time. This would be specifically beneficial in the physical therapy field, where practitioners could examine the walking of their patients throughout their day-to-day activities instead of just during an appointment.

VII. Contribution

We actively collaborated in developing each section, with more focuses outlined as follows:

- Idea Generation: Led by Ehsan, with modifications by Colby.
- Model: Initiated by Ehsan, further developed by Colby.
- Presentation Slides: Prepared by Colby, refined by Ehsan.
- Report: Initiated by Colby and subsequently continued by Ehsan.

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